

Dalron J. Robertson, M.S.

Research Scientist, Pharm.D./MBA Candidate

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Experience Summary

Creator, engineer, and researcher working across the life sciences with deep expertise in biomedical science, clinical pharmacology, and software engineering. Leveraging high performance computing, machine learning, and pipeline engineering, I develop intelligent systems that accelerate discovery, automate workflows, and give structure to meaningful scientific work.

Technical Skills

AI & Large Language Models: ChatGPT | Deepseek | FAISS | Gemini | LLaMa | LoRA | OpenAI | TinyLlama | TxGemma

Bioinformatics & Molecular Modeling: AlphaFold | ChimeraX | SwissTargetPrediction | Molinspiration | UniProt

DataScience & Machine Learning: Tensorflow | Scikit-Learn | Pandas | NumPy | PyTorch

Languages: Javascript | Python | Pinescript | C | PostgreSQL | MySQL

Tools & Technologies: Git | AWS | ISOLIDE | Schrödinger | Jest | DocuTrack | Framework LTC | UiPath

Web Development: Node.js | React.js | Express.js | FastAPI | Flask | HTML | CSS | AJAX | DOM | jQuery

Education

Chicago State University – *Doctor of Pharmacy, Expected May '28*

Mississippi College – *MBA, Expected May '28*

Mississippi College – *M.S., in Biological Sciences*

University of Illinois, Chicago – *Software Development Certificate*

Jackson State University – *B.S., in Biology*

Research & Publications

Restructuring Our Biomedical Knowledge Model with TxGemma & Improved RAG | June 2025

Redesigned the biomedical question-answering workflow by integrating the TxGemma 2B language model and a rebuilt retrieval system, producing more accurate and continuously updated responses on a single Mac mini.

- Replaced the original model with TxGemma 2B, reducing training time from several days to a few hours while improving accuracy.
- Applied lightweight LoRA adapters that reached target loss in about 300 steps, lowering error from 5.16 to 2.23.
- Enhanced retrieval using 1,024-dimension BGE-large embeddings with two-stage ranking for better context selection.
- Built an automated nightly pipeline that ingests new PubMed papers, embeds them, and rebuilds the index without manual effort.
- Automated nightly ingestion of new PubMed papers and index rebuilding.
- Added strict context validation and n-gram checks, reducing unsupported statements by roughly one-third.
- Implemented version-controlled configs, structured logs, and privacy guards to ensure reliable scaling.

The NaS Cortex: A Knowledge Architecture for the Life Sciences | May 2025

Designing a neuro-inspired AI system modeled after human cortical organization to accelerate discovery across biology and medicine.

- **Neuro-inspired Modular Architecture:** Applied cortical principles (Brodmann's areas) to organize and structure domain-specific knowledge efficiently.
- **Specialized Expert Models ("Gyri"):** Built Mixture of Experts models for distinct biomedical fields, enabling deep specialization and rapid retrieval.
- **Central Knowledge Layer for Agentic Systems:** Created a shared substrate that allows agentic systems to query and synthesize knowledge across gyri for complex reasoning.
- **Autonomous & Continuous Learning:** Developed self-updating loops that ingest new literature and retrain specialized models without manual intervention.
- **Dynamic Routing & Integration:** Implemented intelligent routing that fuses outputs from multiple gyri to handle multidimensional scientific problems.
- **Prototype Design and Evaluation:** Led prototyping and benchmarking on Apple Silicon, identifying constraints and next steps for scaling distributed training.

Bioinformatic Profiling and Pharmacological Evaluation of Phytochemical Constituents *N. sativa*

Mississippi College, Dr. Angela W. Reiken | Jan 2025 – May 2025

<https://bit.ly/DJR-PhytochemicalResearch>

*Conducted comprehensive research investigating the pharmacological and therapeutic potential of ethanol-derived *Nigella sativa* extracts using bioinformatics, computational chemistry, molecular modeling, and experimental validation. Presented findings at the 24th Annual Graduate Research Forum and awarded 2nd Place – Foman & Nita Musselwhite Prizes, recognized by the Provost for excellence in research pipeline and methodology.*

- **Computational Pipeline:** Built a compound library, predicted protein targets with SwissTargetPrediction, modeled 3D structures via AlphaFold, and conducted molecular docking using Schrödinger Suite and CB-Dock2.
- **Analytical and Experimental Work:** Performed antioxidant and anti-inflammatory assays, quantified flavonoid content, and analyzed results through spectrophotometric methods.
- **Mentorship and Training:** Supervised and trained a 16-member research team in bioinformatics, molecular modeling, and visualization using AlphaFold, ChimeraX, and Blender.

Comparative Enzymatic Hydrolysis of Lactose Across Milk Types Using Exogenous Lactase

Mississippi College, Dr. Angela W. Reiken | Sep 2024 – Dec 2024

Designed and conducted an original experiment evaluating the effectiveness of exogenous lactase across milk types with varying fat content using glucose-based colorimetric assays to quantify lactose hydrolysis and enzyme efficiency.

- **Experimental Design:** Simulated lactose digestion across five milk conditions (whole, 2%, 1%, fat-free, and almond) to assess enzyme functionality and fat interference.
- **Data Collection and Analysis:** Measured glucose pre- and post-treatment using semi-quantitative test strips at multiple time points and analyzed results through enzyme kinetics to determine catalytic efficiency.
- **Findings and Presentation:** Observed consistent enzymatic performance with minimal fat impact on hydrolysis rate. Presented results to faculty and peers, highlighting methodological limits, clinical implications, and future research directions.

Experience

UChicago Medicine, Hyde Park, Chicago, IL

Pharmacy Intern — Discharge Medication Program | March 2026 -

- Work across the Discharge Medication Program and outpatient pharmacy, preparing and dispensing prescriptions, coordinating bedside delivery, and supporting continuity of medication therapy after hospitalization.
- Prepare a broad and varied range of nonsterile compounded medications across oral and topical dosage forms, applying established formulas, pharmaceutical calculations, documentation standards, and quality-control procedures to support individualized patient needs.
- Operate Parata Max automated dispensing systems, including prescription filling, stocking, replenishment, workflow monitoring, and troubleshooting; assist with receiving, reconciling, and ordering pharmacy inventory.
- Conduct insurance, benefits, and financial investigations; complete and track prior authorizations; communicate with insurers; and escalate denials or delayed responses to pharmacists and clinical teams.
- Facilitate enrollment in manufacturer and foundation patient-assistance programs, collaborating with social services and case management to connect eligible patients with financial support.
- Document medication-access activity in pharmacy tracking systems, manage inbound and outbound patient calls, monitor approval status, and identify recurring reimbursement or workflow barriers.
- Review medication histories and patient profiles, support medication reconciliation and discharge counseling, and communicate medication-related concerns to pharmacists and the multidisciplinary care team under supervision.

Astellas Pharmaceuticals, Northbrook, IL

Mentee - Bridge to Industry Intensive Program | Oct 2025 -

- Selected participant in a professional development partnership between Chicago State University College of Pharmacy and Astellas Pharma Global Development.
- Engaging in one-on-one mentorship with Astellas executives, gaining insights into Quality, Research, and Regulatory leadership.
- Participating in seminars and applied learning sessions focused on pharmaceutical innovation, compliance, and global development pathways.
- Building industry knowledge bridging Pharm.D. training with real-world applications in biotechnology and drug development.

Walgreens, Chicago, IL

Pharmacy Intern | Oct 2025 - Jul 2026

- Participate in supervised pharmacy practice, supporting medication verification, dispensing workflows, and patient care in accordance with state and federal pharmacy regulations.
- Evaluate patient medication profiles and prescription data using pharmacy information systems to support accuracy, appropriateness, and regulatory compliance.

Mississippi College, Clinton MS

Lead Graduate Researcher | Dec 2024 - Jul 2025

- Employed OpenAI LLMs to extract and synthesize bioactivity data, phytochemical profiles, and pharmacological properties from PubMed, PubChem, and UniProt.
- Built a curated *Nigella sativa* compound library, performed molecular descriptor and drug-likeness screening (SwissADME, Molinspiration), and prioritized leads using PCA and clustering.

- Predicted protein targets with SwissTargetPrediction, modeled 3D structures in AlphaFold, and conducted molecular docking with Schrödinger Suite and CB-Dock2.
- Performed antioxidant, anti-inflammatory, and phytochemical assays to validate computational findings.
- Trained and supervised 16 undergraduate and graduate researchers in bioinformatics and modeling.

Scale AI, San Francisco, CA

AI Software Engineer | Sept 2023 - Jan 2025

- Collaborated with Google to enhance the Gemini large language model through optimized dataset curation and evaluation design.
- Improved YouTube and Google Search integration by enabling multimodal models to process video content and deliver more contextually relevant search outputs.
- Applied Side-by-Side (SxS) evaluation and prompt-based analysis to refine model accuracy and response quality.
- Developed synthesis algorithms to generate temporal and relational training data for model understanding.
- Conducted red-teaming analyses to identify and mitigate model vulnerabilities.

Author & Self-Publisher

Independent | Aug 2023 - Present

- Published three full-length titles: *Circuit Breakers: Echoes of Innovation*, *Circuit Breakers: Imperfections*, and *To Love Ignorantly: Love's Memoir*.
- Managed end-to-end publishing, including manuscript development, editing, design, formatting, and distribution through Amazon KDP, Google Books, and Barnes & Noble.
- Developed marketing and metadata strategies to optimize visibility and ensure quality across digital and print editions.

Medication Management Partners, Crestwood, IL

Pharmacy Informatics & Automation, Pharmacy Intern | Aug 2020 - Apr 2022

- Managed prescription workflows for 150+ LTC facilities, including verification (QV1/QV2), medication review, and DURs.
- Led daily standup meetings and participated in weekly department head sessions, offering system-level insights and operational feedback to support organizational performance.
- Served as informatics lead for Framework LTC, DocuTrack, and eMAR systems, ensuring script accuracy and seamless provider communication.
- Operated and maintained TCGRx ATP, Parata Max, and IntelliVault robotics platforms to support accurate dispensing and compliance.
- Collaborated with providers and insurers to resolve prior authorizations, adjudication errors, and patient-specific issues.
- Trained new staff across automation, informatics, and clinical workflows in a fast-paced startup environment.

CVS Health, Chicago, IL

Pharmacy Intern | Jul 2015 - Aug 2020

- Supported daily prescription processing and workflow coordination under direct pharmacist supervision
- Assisted with patient counseling, medication dispensing, and insurance claim resolution.

Oral Presentations

“Atrial Fibrillation: Mechanisms, Thromboembolic Risk, and Contemporary Pharmacologic Therapy”

Pharmacy Patient Skills III | Chicago State University, Nov 2025

Presented an evidence-based review of atrial fibrillation pathophysiology, CHA₂DS₂-VASc risk stratification, and current rate, rhythm, and anticoagulation management strategies..

“Computational Drug Discovery from Nigella sativa: A Multimodal Pipeline Integrating Bioinformatics, Structural Modeling, and AI Architecture”

Mississippi College Research Week | Mississippi College, Apr 2025

Presented a bioinformatics-based drug discovery pipeline using PubChem, SwissTargetPrediction, ChimeraX, and ISOLDE to explore compound–target interactions and ADMET properties.

“Code of Ethics for AI-Driven Biotechnology Leadership”

Ethics of Healthcare Administration | Mississippi College, Mar 2025

Outlined an ethical framework for AI and quantum biotechnology governance, emphasizing patient safety, data privacy, and equitable innovation.

“Breast Cancer Risk, BRCA Mutations, and Targeted Therapies: A Ted Talk on Genomics and Hope”

Biology of Cancer | Mississippi College, Dec 2024

Delivered a TED Talk–style presentation on BRCA mutations, genomic risk, and targeted therapies for breast cancer awareness.

“Artificial Intelligence and Quantum Computing in Drug Discovery and Genomics: Paving the Way for Personalized Medicine”

Graduate Seminar | Mississippi College, Nov 2024

Explored applications of AI-enhanced quantum simulations in precision medicine and biomarker discovery.

Poster Presentations

“Bioinformatic Profiling and Pharmacological Evaluation of Phytochemical Constituents from N. sativa”

Mississippi College Research Week | Mississippi College, Apr 2025

Presented computational and experimental research on *Nigella sativa* extracts using bioinformatics, molecular modeling, and pharmacological validation.

“Comparative Enzymatic Hydrolysis of Lactose Across Milk Types Using Exogenous Lactase”

Mississippi College, Dec 2024

Presented experimental findings on lactase enzyme activity across varying milk fat contents using glucose-based colorimetric assays.

Books & Creative Publications

Circuit Breakers: Imperfections (2025)

Second installment in the *Circuit Breakers* series. Set in the biotech industry, exploring corporate pressure, ethical conflict, and the human cost of innovation.

To Love Ignorantly: Love's Memoir (2025)

A reflective memoir in eight acts examining love, faith, vulnerability, and personal transformation.

Circuit Breakers: Echoes of Innovation (2024)

First novel in the *Circuit Breakers* series. Set in Chicago's medical and technology sectors, addressing AI ethics and humanity in a machine-shaped world.

Awards & Honors

2nd Place - Graduate Research Forum, Foman & Nita Musselwhite Prizes | Mississippi College, 2025

Awarded 2nd place at the 24th Annual Graduate Research Forum for "*Bioinformatic Profiling and Pharmacological Evaluation of Phytochemical Constituents from Nigella sativa*." Recognized for developing a computational framework integrating bioinformatics, 3D protein modeling, and machine learning.

Graduate Academic Excellence - 4.0 GPA (Winter & Spring) | Mississippi College, 2025

Maintained a perfect 4.0 GPA across consecutive semesters in the M.S. Biological Sciences program, reflecting consistent academic and research excellence.

Clerkship/Rotations

Kindred Hospital, Chicago North

Pharmacy Practice IV (IPPE), Jan 2026 - May 2026

Long-term acute care hospital (93 beds) with High Acuity and Respiratory Failure-certified units.

- Reviewed inpatient medication orders for accuracy, dosing, renal adjustments, and drug-drug interactions.
- Participated in sterile compounding activities in USP <797> compliant IV environment, including IVPBs, large-volume parenterals, and TPM order review.
- Performed medication reconciliation and documented pharmacist-guided clinical interventions.
- Utilized BD Pyxis™ automated dispensing cabinets for medication distribution, override review, and controlled substance monitoring
- Responded to drug information inquiries and collaborated with nursing and physician teams during patient care discussions and rounding.

Drexler Pharmacy, Chicago IL

Pharmacy Practice III (IPPE), Aug 2021 - Dec 2021

Independent community pharmacy providing direct patient care and chronic disease medication management within a neighborhood-based setting.

- Assisted with prescription processing, verification, and patient counseling under pharmacist supervision.
- Provided medication education for chronic conditions including hypertension, diabetes, hyperlipidemia, and anticoagulation therapy.
- Conducted medication reconciliation and addressed adherence concerns through direct patient interaction.
- Collaborated with prescribers' offices regarding clarifications, refill authorizations, and therapy modifications.

Harbour Light Hospice, Glen Ellyn, IL

Pharmacy Practice II (IPPE), Jan 2020 - March 2020

Outpatient hospice setting providing interdisciplinary end-of-life care during the COVID-19 pandemic.

- Collaborated with nursing staff, occupational therapy, and healthcare administration to support comfort-focused medication management.
- Reviewed and discussed pharmacotherapy for symptom control, including pain management, anxiolysis, dyspnea, and end-of-life comfort measures
- Participated in interdisciplinary care discussions addressing medication appropriateness, deprescribing, and quality-of-life considerations
- Observed care coordination processes within a hospice model emphasizing dignity, symptom relief, and family support

Walgreens Pharmacy, Orland, IL

Pharmacy Practice I (IPPE), Aug 2019 - Dec 2019

Community pharmacy introduction to retail pharmacy operations and patient service.

- Observed and assisted in prescription processing, medication dispensing workflow, and third-party adjudication procedures.
- Participated in basic patient counseling activities under pharmacist supervision
- Gained exposure to community pharmacy regulations, inventory management, and controlled substance handling protocols

Interprofessional Clinical Training

UChicago Medicine

Sep 2020 - May 2021

Completed a seminar-based program on cross-disciplinary collaboration and healthcare delivery. Worked in rotating teams to analyze specialty-specific patient cases and develop team-based care solutions.

Rush University Medical Center

Sep 2019 - May 2020

Engaged in weekly interdisciplinary training alongside students from medicine, pharmacy, nursing, physician assistant studies, and physical therapy. Participated in collaborative treatment planning, ethics discussions, and supervised patient rounds emphasizing coordinated care and communication.

Certifications

American Red Cross - *Adult First Aid/CPR/AED Certification*

American Pharmacists Association (APhA) - *Pharmacy Based Immunization Delivery Certification*

Genesis Plus Trading Academy

National Healthcareer Association (NHA) - *Certified Pharmacy Technician (CPhT)*

Pharmacy Technician Certification Board (PTCB) - *Certified Pharmacy Technician (CPhT)*

Licensure

Illinois Department of Financial and Professional Regulation (IDFPR)

Pharmacy Intern - License No. 049266668 | ACTIVE

Leadership & Governance

Student National Pharmaceutical Association (SNPhA)

Fundraising Chair, Executive Board, Dec 2025 - Present

- Serve on the Executive Board, contributing to chapter governance and strategic planning.
- Lead fundraising initiatives to support conferences, outreach, and professional development.
- Develop sponsorship outreach and donor engagement strategies.

Chicago State /university, School of Pharmacy

Student Ambassador - Curriculum & Assessment Committee, Oct 2025 - Present

- Represent the student body during the active accreditation review cycle.
- Participate in standards review, curriculum assessment, and continuous quality improvement discussions.
- Collaborate with dean facility leadership in accreditation preparation meetings.
- Contribute to documentation review and consultant-facing discussions

Professional Organizations

Sep 2019 - Present

American Pharmacist Association Academy of Student Pharmacists - (APhA - ASA)

Sep 2019 - Present

Student National Pharmaceutical Association - (SNPhA)

Community Service

Volunteer

Hot Chocolate 15K/5K Run

Chicago, IL

Volunteer

Chicago Heart Walk

Chicago, IL

Volunteer

University of Mississippi Medical Center - Supervisor: Marsha Burton

Jackson, MS